

claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_f10e69b6-333\agent-af6f54ea41a3f3997.jsonl`  
- SHA-256: `6a3fb216bbe26f77e906190db25317008c3175c1185934f461ae78a1a405a56e`  
- Source modified: `2026-06-15T14:59:22+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_f10e69b6-333`  
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`
```

Transcript

1. user (2026-06-15T14:57:50.099Z)

You are reviewing a DRAFT product-overview document for "KhelSutra" ? a local, AI badminton highlight/rally tool.
The draft lives at C:/Users/avidu/Projects/badminton-highlight-indexer/docs/KHELSUTRA_PRODUCT_OVERVIEW.md (Read it). It is dual-audience: (a) a non-technical FIELD ASSOCIATE who will talk to academies, and (b) invited ALPHA/BETA academies/coaches/players who will use the tool and give feedback. It must be PRODUCT-LEVEL (no fine-grained technical detail) and HONEST (no overpromising to real prospective customers).

GROUND TRUTH (from the repo's docs/memory ? what is actually real):

- TODAY the tool genuinely does: upload a long video ? automatic rally detection ? highlight reel (with a "Generated by KhelSutra.guru" watermark) ? a simple summary (rally count + total play time); re-process/download;
private local-first web app (async jobs + chunked upload + highlight download are live & E2E-verified); runs on real 4K GoPro match footage.
- Quality reality (DO NOT put raw numbers in a customer doc): clean single-court footage works very well (~0.93 F1);
busy MULTI-COURT footage is markedly harder (~0.66) and is the explicit moat/focus. The doc should convey this qualitatively ("very good on clean single-court; multi-court is harder and is what we're improving").
- ROADMAP (NOT today): shot detection (smash/drop/serve), deeper per-rally/per-match analysis, per-player history,
"AI assistant coach", more sports (table tennis next), Kannada/Hindi localization.
- THE FLYWHEEL (the headline the owner wants foregrounded): players/coaches can run their OWN videos; coaches can
correct what it found; over time + more videos the model sharpens AND that library unlocks the deeper analysis.
- GUARDRAILS (hard, non-negotiable, must be represented accurately): using a user's video to improve/train the model
is OPT-IN (explicit permission), never automatic; footage involving MINORS is NEVER used for model improvement;
private/local-first ("your video, your data"); pricing is NOT set.
- Capture quality is the gate: fixed camera behind court, full court in frame, landscape, steady, decent light;
handheld/panning/broadcast footage is the failure mode.
- It is NOT line-calling/scoring/umpiring, NOT player biometrics, NOT a coach replacement.

RULES: read-only; you cannot edit. Return STRUCTURED output (consumed programmatically). Quote the draft line/phrase when flagging. Be specific and actionable. Do not invent product capabilities.

LENS: CONSENT / PRIVACY / SAFETY. Your job: verify every privacy & consent claim is accurate and prominent enough for a customer-facing doc. Required, accurately stated: opt-in (never automatic) to use a user's video for model improvement; minors' footage NEVER used for model improvement; private/local-first; we don't publish footage. Flag any phrasing that could be read as "we train on your video by default" or that under-states the minors rule. Also flag any data claim we cannot stand behind. This is the highest-stakes lens ? be strict.

2. assistant (2026-06-15T14:57:52.840Z)

I'll review the draft document through the consent/privacy/safety lens. Let me start by reading it, then cross-reference against the ground truth in the repo's docs/memory.

3. user (2026-06-15T14:57:53.744Z)

```
1      # KhelSutra ? Product Overview & Early-User Guide
2
3      *What it does today, how to get the best out of it, and where it's going.*
4      *For field associates and for invited alpha / beta academies, coaches, and players.*
5
6      ---
7
8      ## In one line
9
10     **KhelSutra turns ordinary badminton video into useful pieces ? automatically.** Point a
phone or
11     GoPro at the court, record a session, and KhelSutra finds every rally, stitches a
**highlight reel**,
12     and gives you a **simple summary** (how many rallies, how much real play time). No
special hardware,
13     no editing, no manual scrubbing.
14
15     ---
16
17     ## The problem we're solving
18
19     Academies and serious players already record a lot of video ? and then almost never
watch it. Hours
20     of footage pile up on phones and memory cards because cutting it into useful clips by
hand is tedious
21     and nobody has the time. The good rallies, the moments worth reviewing, stay buried.
22
23     And most academy halls run **several courts at once**, so a single recording is a busy,
noisy scene.
24     Ordinary video tools choke on that. **Pulling the real rallies out of that footage ?
automatically ?
25     is exactly what KhelSutra is built to do.**
26
27     ---
28
29     ## Who it's for
30
31     - **Coaching academies & coaches** ? turn raw session footage into review-ready clips
and reels for
32     teaching, without an editor.
33     - **Serious players** ? get clips of your own rallies to review, keep, and share.
34     - Today's focus is **badminton academies in Bengaluru** ? that's who we're building with
first.
35
36     ---
37
38     ## What it does **today**
39
```

40 This is what's working right now, in the private web app invited users can try:

41

42 - ****Upload a full session or match video**** (from a phone on a tripod, or a GoPro). Long videos are fine.

43 - ****Automatic rally detection**** ? it finds where each rally starts and ends, on its own.

44 - ****Highlight reel**** ? the rallies stitched into one clip you can watch and share, with a small

45 **"Generated by KhelSutra.guru"** mark.

46 - ****A simple summary**** ? how many rallies were found and how much actual play time there was.

47 - ****Re-use & download**** ? process once, come back to it, download the reel.

48

49 That's the core loop: ****record ? upload ? get back a reel + summary.**** No editing skill needed.

50

51 ---

52

53 **## How to get the best out of it (capture matters)**

54

55 KhelSutra works best on a ****steady, full-court recording**** ? the same way a coach would set a camera

56 to review play. A few simple habits make a big difference to the results:

57

58 - ****Put the camera behind the court, slightly elevated****, with the ****whole court in frame****.

59 - ****Keep it still**** ? a fixed tripod beats a handheld or panning shot every time.

60 - ****Shoot landscape****, with reasonable lighting.

61 - ****Record the whole session**** and upload it as-is ? KhelSutra does the cutting.

62

63 Footage that fights these habits (shaky handheld, heavy zooming/panning, broadcast-style cuts, very

64 dark halls) is the ****hardest case**** and where results suffer most. When in doubt: fixed camera, full

65 court, don't touch it.

66

67 > A one-page capture checklist is provided to academies in the trial.

68

69 ---

70

71 **## The part that makes KhelSutra different: **it learns from your videos****

72

73 This is the heart of the product, and what sets it apart from a generic "AI clipper":

74

75 ****Your videos make _your_ tool sharper.**** As you and your players run more of your own footage through

76 KhelSutra ? and as coaches confirm or correct what it found ? the underlying model gets better at

77 ***your* courts, *your* lighting, *your* players, and the busy multi-court halls that trip up everything**

78 else. It improves with use instead of staying frozen.

79

80 That same growing library of real, labelled footage is also what unlocks the ****deeper features below**** ?

81 which is why early users aren't just customers, they're co-builders. Every session you contribute makes

82 the next version smarter.

83

84 ****Your data stays yours.**** Using your video to improve the model is ****opt-in**** ? nothing is used to train

85 anything without your explicit permission. We don't publish anyone's footage. ****Sessions involving minors**

86 are never used for model improvement.** The product is built ****private and local-first****: your video, your

87 data, your control.

88

```

89     ---
90
91     ## Where it's going ? toward an **AI assistant coach**
92
93     Today KhelSutra finds rallies and makes reels. The roadmap ? fed by the growing video
94     library above ?
95     turns it into a genuine training companion:
96
97     - **Shot detection** ? recognising smashes, drops, net shots, serves, and clears within
98     each rally.
99     - **Deeper per-rally and per-match analysis** ? patterns, tendencies, what's working and
100     what isn't.
101     - **Per-player & per-match history** ? track a player's progress over weeks and months,
102     not just one clip.
103     - **Personalised insights** ? an **AI assistant coach** that supports (never replaces)
104     the real coach,
105     surfacing what to work on next.
106     - **More sports** ? table tennis next, then other racket/court sports.
107     - **Your language** ? Kannada and Hindi, alongside English.
108
109     > **Read this honestly:** the items in this section are the **direction**, not today's
110     buttons. What you
111     > can use *today* is rally detection + highlight reels + the play-time summary. The
112     deeper analysis arrives
113     > as the video library grows ? which is exactly why your footage and feedback matter so
114     much.
115
116     ---
117
118     ## What we're focused on over the next ~4 weeks
119
120     1. **Growing the library of real academy footage** ? especially busy **multi-court**
121     halls (the hard,
122     valuable case). This is the single biggest lever on quality.
123     2. **Field conversations with academies** ? understanding how coaches and players
124     actually work with
125     video, and what would make this genuinely useful.
126     3. **Recording sessions and showing academies their own footage** as reels ? "this is
127     what it looks
128     like for you."
129     4. **Sharpening multi-court accuracy** and inviting a few more academies into the
130     private trial.
131
132     ---
133
134     ## What KhelSutra is **not** (so expectations stay honest)
135
136     - It's an **early, invite-only trial** ? useful, but still improving; you'll hit rough
137     edges.
138     - It is **not a line-calling, scoring, or umpiring system** ? it finds rallies and
139     highlights.
140     - **Shot detection and the deeper "AI coach" analytics are on the roadmap, not available
141     today.**
142     - It is **not a replacement for a coach** ? it's a tool that saves coaches time and adds
143     insight.
144     - Accuracy depends on **capture quality** ; messy handheld/broadcast footage is a known
145     weak spot.
146     - **Pricing isn't set** ? we're still learning what's fair; today's invited use is about
147     getting it right.
148
149     ---
150
151     ## Quick answers to common questions
152
153     ***"How accurate is it?"***

```

136 Very good on a clean, single-court recording. A busy multi-court hall is harder ? and is exactly what

137 we're improving. The best answer is to *show* a sample reel rather than quote a number.

138

139 ***"Do I need a special camera or hardware?"**

140 No. A phone or GoPro on a tripod, behind the court, whole court in frame. We provide a one-page checklist.

141

142 ***"Is my video private? What happens to it?"**

143 It's built private and local-first ? your video stays yours. We never use anyone's footage to improve the

144 model without explicit opt-in, we don't publish footage, and **footage involving minors is never used**.

145

146 ***"Can it tell me the score / who won / call lines?"**

147 Not today. Today it finds rallies and builds a highlight reel + a play-time summary.

148

149 ***"Does it work for doubles and tournaments?"**

150 Yes for finding rallies. Crowded multi-court tournament halls are the hard case we're specifically working on.

151

152 ***"Will it get better?"**

153 Yes ? that's the whole idea. The more real footage runs through it (with permission) and the more coaches

154 correct it, the sharper it gets, and the sooner shot detection and deeper analysis arrive.

155

156 ***"Which sports?"**

157 Badminton today; table tennis and other racket sports are on the roadmap.

158

159 ***"When can I use it / how do I join?"**

160 It's an invite-only private trial. If you're interested, the best next step is to let us record a session

161 and show you your own court and players as a reel.

162

163 ---

164

165 *KhelSutra · khelsutra.guru · Internal & invited-trial overview, June 2026.*

166 *Field associates: the conversation script and visit rules live in the separate Field Visit Playbook.*

167

4. assistant (2026-06-15T14:57:57.338Z)

Now let me cross-reference against the ground truth. Let me check the repo's authoritative privacy/consent docs and memory to verify each claim, and confirm there's nothing in the actual product that contradicts the privacy posture (e.g., the watermark/cloud-Gemini interaction).

5. user (2026-06-15T14:57:59.255Z)

1 # CLAUDE.md ? project memory / agent handoff

2

3 Local, AI-powered **badminton (? racket-sports) highlight indexer + rally detector**:

4 FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid **Gemini** as the cloud AI. Sibling

5 repos: `sports-data-collector`, a VLC-based `rally-annotator`.

6

7 ## Read these first

8 - **`docs/README.md`** ? doc index. **`docs/CODE\_MAP.md`** ? code navigation + data

9 contracts (the cold-start map; read before touching code).

10 - **Authoritative owned-model roadmap:** `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` (+

the

11 live `docs/NEXT\_STEPS.md`). These SUPERSEDE the 2026-06-07 strategy docs below where

they

12 disagree (licensing, Gemma-4 stance, base model, conversational-QA).

13 - **Strategy (post-scaffolding):** `docs/COMMERCIALIZATION.md`,

```

`docs/PMF_MARKET_RESEARCH.md`,
14     `docs/PLATFORM_ARCHITECTURE.md`, `docs/DEFERRED_HARDENING_PLAN.md`. Also
`docs/ROADMAP.md`
15     (offline-segmenter narrative) and `docs/BACKLOG.md`.
16     - Rally-detection quality track (design merged 2026-06-13, owner-gated, NOT
started):**
17     `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model) ?
`docs/MULTI_SIGNAL_FUSION_PLAN.md`
18     (fuse shuttle + person + audio cues; the held-shuttle bug is the already-built
`rally_gate.py`
19     "Gate A" not wired into production ? cheapest win) ? `docs/EXPERIMENT_HARNESS.md`
(A/B + shadow
20     eval so cues land flag-gated/default-OFF with zero regression) ?
`docs/ROORA_LABELING_GUIDELINES.md`
21     (v8 vendor labeling spec). docs/NEXT_STEPS.md has the prioritized pick-up for this
track.**
22     - History: `docs/archives/` ? ADRs (`decisions/DECISIONS.md`), research, and
23     checkpoints/` (latest: `2026-06-strategy-foundation/`). Archive policy: only move
24     terminal-state (DONE/REJECTED) work; live docs stay at `docs/` top level.
25
26     ## Current state (June 2026)
27     - Scaffolding complete; strategy foundation + owned-model roadmap merged. No
28     platform/owned-model implementation has started. A 2026-06-10 audit-driven hardening
29     sweep landed (licensing gate, detector keying, re-ingest safety, config, eval gate,
API
30     validation, reliability, CI + test coverage) ? see git history.
31     - Two tracks, both owner-gated: the committed next PLATFORM slice = async job
model**
32     (`DEFERRED_HARDENING_PLAN.md` #16, single-tenant); the owned-model track starts at
a
33     greenlit Phase-0 milestone (`OWNED_MODEL_IMPLEMENTATION_PLAN.md`). Do not start
34     implementation without explicit owner approval.**
35
36     ## Architecture in one breath
37     - Pluggable registries (ABC + `Registry` singleton): segmenters / providers /
validators
38     / sports / annotations ? see `backend/pipeline/segmenters/base.py`. Future:
39     `StorageBackend` + `ComputeBackend` registries (PLATFORM $3).
40     - `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one
helper).
41     - Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
42     user_version` migration ladder in `backend/infrastructure/database.py`.
43
44     ## Committed guardrails (non-negotiable)
45     - Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
46     training data source** (terms/copyright). Commercial-clean licensing for EVERY
dependency ?
47     code, data, AND weights: MIT / Apache-2.0 / BSD only (ratified 2026-06-09, owner-
agreed).
48     - Owned-model base: Qwen3-VL (Apache-2.0)** primary, InternVL3 (MIT) fallback;
Gemma-4
49     is AMBER / bench-only (Apache grant likely but Prohibited-Use posture unconfirmed ?
counsel
50     needed; do NOT build on it without sign-off). Reject Gemma 3 (viral license).
Exclude
51     minors** from the training pool; per-user training data needs a separate explicit opt-
in.
52     - v1 = rally + highlight + history + correction loop; gait/biometrics strictly
future**
53     (GDPR Art. 9).
54     - Generic data schema; `coach ? athlete` is an optional overlay, not baked in.
55
56     ## Workflow conventions
57     - Branch off `master`; ONE PR per work item; NO PR stacking. Don't open a PR unless
asked.

```

58 - Tests: `python run\_all\_tests.py` (offline, fully mocked). Python 3.8+; `ffmpeg`
required.
59 - Never push to the default branch directly; descriptive commit messages.
60 - ?? This dev container lacks GPU / torch / cv2 / numpy ? the **full pipeline can't run
here**.
61 Verify pure-logic paths and hand machine-side checks (real video, GPU/WSL suite) to
the owner.
62

6. user (2026-06-15T14:57:59.551Z)

Found 49 files

docs\KHELSUTRA_PRODUCT_OVERVIEW.md
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
docs\README.md
docs\ROADMAP.md
docs\ROORA_LABELING_GUIDELINES.md
docs\TRACKNET_WSL_SETUP.md
docs\PLATFORM_ARCHITECTURE.md
docs\PERSONALIZATION_PLAN.md
docs\ANALYZERS_AND_RECONCILERS.md
docs\BACKLOG.md
docs\CODE_MAP.md
docs\COMMERCIALIZATION.md
docs\DEFERRED_HARDENING_PLAN.md
docs\EVALUATION.md
docs\EXPERIMENT_HARNESS.md
docs\GOLDEN_DATA_SHARING.md
docs\MULTI_SIGNAL_FUSION_PLAN.md
docs\NEXT_STEPS.md
docs\QUALITY_ITERATIONS.md
docs\archives\quality-iterations\README.md
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md
docs\STORAGE_SHARING_MODEL.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\GOLDEN_SET_VENDORING.md
docs\VIDEO_LOCALITY_MODEL.md
docs\PRODUCT_FIELD_ASSOCIATE_FLYER.md
docs\PMF_FIELD_SIGNALS.md
docs\I18N_PLAN.md
docs\FIELD_VISIT_PLAYBOOK.md
docs\UI_ALPHA_EXECUTION_PLAN.md
docs\archives\TEST_RUN_SUMMARY.md
docs\mockups\ui_alpha_mockup.html
docs\archives\decisions\DECISIONS.md
docs\UI_PROPOSAL.md
docs\HOSTING_PLAN.md
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\PMF_MARKET_RESEARCH.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md

7. assistant (2026-06-15T14:58:04.628Z)

Important nuance surfaced: the project uses paid **Gemini** as the cloud AI** for inference. This

means video data may leave the device and go to Google's cloud ? which directly bears on the doc's "private and local-first" and "your video stays yours" claims. Let me verify how Gemini is actually used in the current product (is it on the live upload?reel path, or future/optional), and check the personalization/storage-sharing docs that govern the opt-in and minors rules.

8. user (2026-06-15T14:58:05.601Z)

```
1      # Personalization Plan ? a per-user feature on a generalization-first baseline
2
3      **Status: DESIGN ? owner-adopted (hybrid); Phase-0 engineering landing.** This is the
authoritative
4      plan for the personalization track. The detailed strategy analysis + the full pros/cons
that led here
5      live in the working artifacts `scratch/PERSONALIZATION_PIVOT_ANALYSIS.md` (the decision-
grade
6      counter-case) and `scratch/PERSONALIZATION_FEATURE_DESIGN.md` (the code-grounded
design); this doc is
7      the checked-in summary + decisions + phased build plan. Peer to
8      [OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_PLAN.md),
9      [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md), and
[MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md).
10
11     ## 0. Thesis & strategic rationale (why *hybrid*, not personalization-first)
12
13     **Foundation principle (owner):** "a tool that gets better the more you help it."
Contribution
14     (data / video / diagnostics / annotation) improves BOTH the user's own model AND the
shared baseline;
15     **payoff scales with contribution.**
16
17     A directional pivot ? *personalization-first* (ship a baseline that continually self-
optimizes to one
18     user's own footage) ? was researched in depth. The verdict was **HYBRID, not all-in**:
19
20     - **Adopt personalization as a flag-gated FEATURE on a generalization-first baseline.**
The harness
21     genuinely makes per-user *tuning* near-free (it is already a corpus-relative A/B
engine; a per-user
22     corpus is just a special case). This is real and worth shipping.
23     - **Do NOT bet the strategy on per-user *learning*.** Per-user A/B is the n?6 small-
sample regime
24     *worse* (sharding data when you should pool); the differentiated parts (a learned per-
user model
25     serving path, the correction?label write path, a user/tenant dimension) are largely
unbuilt; a
26     naively-overfit per-user model loses the cross-user data flywheel and has no shared
baseline to
27     regress against.
28     - **The owner's contribution-flywheel + free-tier-pooling closes the one real hole** the
counter-case
29     found: by having (free-tier) users contribute to the *shared* baseline, the
compounding cross-user
30     moat is preserved *alongside* personalization ? personalization **and** flywheel, not
either/or.
31
32     Net: a generalization-first **baseline** ("batteries"), a thin **per-user tuning** layer
gated by the
33     honest small-N statistics (with a hard floor + structural-guard protection), and a
**contribution
34     loop** that feeds both.
35
36     ## 1. Owner decisions (locked 2026-06-14)
37
38     1. **Identity** ? auth WILL land; *design for the north star (multi-user / cloud),
implement for
```



```

39         current use (single-user local)*. `user_id` nullable now (NULL = local install),
device/install
40         UUID for the alpha; real auth + a `users` table later.
41     2. **Per-user model storage** ? inline `classifier_json` in the (local) DB now, but
**behind a load
42     interface** so that post-auth it is fetched from the **cloud-hosted DB run by the
service** that
43     generates base models and orchestrates the product. The local store is one
implementation of that
44     interface.
45     3. **`min_n` is a PROMOTION-confidence floor, NOT a service/segmentation gate.** A
brand-new user with
46     one video is ALWAYS fully segmented by the baseline batteries ? never denied, never
"give me more
47     videos first." `min_n` only gates whether a *per-user* A/B is trusted enough to
OVERRIDE a baseline
48     cue default *for that user*; below it (or if the CI / sign test don't clear) ? keep
the baseline.
49     **Training-corpus policy:** *no more than needed, no less than all-if-few* ? fit on
the
50     `smallest_k_with_lift` videos (the held-out-USER learning curve), else ALL available.
51     4. **"Flip Gate-A to default" = ship the validated Gate-A held-shuttle guard as the
**default
52     detection path** (today it lives in the opt-in `FusionSegmenter`, which is not the
default
53     segmenter). Changes served behavior ? **gated on a golden-set regression run + owner
OK**; still
54     open (see §5).
55     5. **Flywheel = backend property** (not in-product marketing). A **free tier** is likely
and
56     **free-tier videos ARE pooled** into the baseline (consent baked into the free terms;
57     minors-exclusion still enforced) ? the clean value-exchange that feeds the owner-
trained base model.
58     Paid tiers get stronger privacy / opt-out.
59
60     **Pending sub-decision:** structural-guard scope ? recommended **Gate-A only** (the
proven structural
61     held-shuttle guard; its value is in failure modes a user's corpus may not yet contain),
with **Gate-B**
62     (person-occupancy / `min_players`) left *subject* to per-user A/B; **`min_n` = 5`**.
**Gate-A-only
63     structural, `min_n` = 5` ? owner-confirmed 2026-06-14.**
64
65     ## 2. Architecture
66
67     **Two-layer decoupling (already real).** Detection (GPU, run once): `FusionSegmenter`
persists the
68     **full superset** of per-window features into `candidate_segments.stats`; only the
served handoff is
69     gated. Decision (CPU, swappable): `experiment.Variant` + `predict()` re-score persisted
features.
70     **Personalization == a per-user `Variant`** (cue_flags + `min_crossings`/`min_players` +
optional
71     LogReg + threshold). No segmenter rewrite ? a per-user `FusionConfig` overlay resolved
upstream.
72
73     **The single highest-leverage wiring:** `fusion_hybrid.py::_select_for_handoff` is the
production
74     decision seam. When a per-user model is pinned AND the promotion gate says ON, apply
75     `LogisticRegression.predict_proba` over `window_stats` (the same math
`experiment.predict()` runs
76     offline). The **load half of the bridge already exists** ?
77     `backend/pipeline/personalization/serving.py::load_user_classifier` rehydrates the
stored
78     `classifier_json` (and `resolve_user_fusion_config` overlays the per-user

```

```

`fusion_config`) from
79     the v5 store ? but it is **not yet wired into the served path**: `_select_for_handoff`
still reads
80     only the static `idx_cfg["fusion"]`. Closing that one seam (the first change to served
behavior, so
81     owner-gated) is the remaining #1 wall.
82
83     **Per-user data model** (migrations follow the `PRAGMA user_version` ladder; current =
v5):
84     - **v4** ? ? **shipped.** Nullable `user_id` on `videos` + `candidate_segments` (NULL =
local;
85     byte-identical to today).
86     - **v5** ? ? **shipped.** `user_models` store: `(user_id, version, baseline_version,
87     feature_schema_hash, fusion_config, classifier_json, promotion_evidence,
n_videos_at_fit,
88     is_active)`. Inline `classifier_json` (tenant-safe); `feature_schema_hash` is the
89     MINOR-vs-MAJOR upgrade switch.
90     - **v6** ? **pending.** `contribution_consent` ledger (default-deny; minors-exclusion
enforced **at
91     the pooling query**, not advisory).
92
93     ## 3. Constructs (the gap list + what is built)
94
95     | # | Construct | Status |
96     |---|---|---|
97     | 1 | Per-user store + serving bridge ? **store + load half built** (v5 `user_models` +
`serving.py::load_user_classifier`); only wiring `_select_for_handoff` remains | Phase 1 (the #1
wall ? load half done, wiring pending) |
98     | 2 | Correction-loop ? per-user labels (`human_label` setter + confirm/reject API;
recall-protecting de-bias prior) | Phase 2 (`human_label` is a read-only stub today) |
99     | 3 | **Per-user promotion gate** (min-N floor + structural-guard exemption) | ? **DONE
? PR #141** (`backend/eval/per_user_gate.py`) |
100    | 4 | Contribution flywheel (opt-in data/video/diagnostics ? shared baseline; payoff ?
contribution; consent + minors-exclusion) | Phase 2 |
101    | 5 | User-as-annotator charter (rally-annotator: auto-generate a ~2-min high-
uncertainty segment from the user's own video; guide labeling ? grounding) | Phase 3 |
102    | 6a | **Held-out-USER learning curve** (the falsifiable "<K videos to superior quality"
criterion) | ? **DONE ? PR #142** (`backend/eval/per_user_eval.py`) |
103    | 6b | Per-user no-regression guard (frozen per-user CONTROL; never accept a regressing
retrain/upgrade) | ? **in flight** (`feat/per-user-no-regression`) |
104    | 7 | Day-0 "batteries" = the validated label-free Gate-A/B fusion (?F1 +0.074, 6/6,
p=0.031) shipped as the default | Phase 0, owner-gated ($5) |
105
106    ## 4. Phased PR-sized build plan + status
107
108    **Phase 0 ? safety + day-0 batteries** (no schema / API / served-behavior change; pure-
Python + tests):
109    - PR-0.1 per-user promotion gate ? ? **#141 merged**
110    - PR-0.2 held-out-USER learning curve ? ? **#142 merged**
111    - PR-0.3 per-user no-regression guard ? ? in flight
112    - PR-0.4 ship Gate-A as the day-0 battery ? **owner-gated** ($5; needs a golden-set
regression run)
113    - PR-0.5 within-user LOVO measurement (the "confirm-or-kill" experiment) ? **owner-
gated** (needs multi-video-per-user data on the GPU box)
114
115    **Phase 1 ? per-user tuning serving + store:** ? already landed ? v4 migration (nullable
`user_id`),
116    v5 `user_models` + CRUD (`upsert_user_model` / `get_active_user_model`), and
`serving.py`
117    (`resolve_user_fusion_config` overlay resolver + `load_user_classifier`). **Remaining:**
`set_user_id`
118    (and the Phase-2 `set_human_label` writer); **wire the bridge into `_select_for_handoff`
(owner-gated
119    ? served path)**; `/retune` + `/model` trust-feature endpoints.
120

```

```

121     **Phase 2 ? correction loop + flywheel:** `POST /api/candidates/{id}/label` (the first
`human_label`
122     writer); recall-protecting de-bias prior; v6 consent ledger + minors-exclusion; **opt-in
contribution
123     upload (owner-gated ? privacy surface)**; free-tier pooling.
124
125     **Phase 3 ? annotator charter + taxonomy + baseline upgrades:** active-learning task
picker; rally-
126     annotator CSV ingest ? per-user labels; **coarse hard/soft shot proxy (owner-gated)**;
**baseline-
127     upgrade governance (MINOR carry / MAJOR re-fit on `feature_schema_hash`) + optional
federated nudge
128     (owner-gated, counsel).**
129
130     ## 5. Open decisions (owner)
131
132     1. **Structural-guard scope + `min_n` value** ? lock **Gate-A-only structural, `min_n =
5`** (recommended)?
133     2. **Gate-A default flip (PR-0.4)** ? Gate-A lives in the non-default `FusionSegmenter`,
so "ship as
134     default" is a path choice (make that segmenter the default vs port the gate into the
default path).
135     Recommended sequence: **run a golden-set regression first**, bring the number, *then*
approve.
136     3. **Per-user `user_id` for the alpha** ? device/install UUID acceptable, or must auth
land first?
137     4. **Local-install flywheel default** ? fully-local zero-pooling until consent is
explicitly granted?
138
139     ## 6. Honest risks (carried from the analysis)
140
141     - **Small-N per-user eval noise** is central and *worse* per-user ? mitigated
structurally: the gate
142     refuses below `min_n` and falls back to the baseline; never auto-promotes in the
regime the tooling
143     exists to refuse in. PR-0.5 confirms-or-kills the core premise before any serving
build-out.
144     - **Cold-start** ? the day-0 Gate-A battery (zero per-user labels) is the floor for the
median user.
145     - **Privacy deferral** ? do not advertise "data never leaves your machine" until on-
device ships;
146     near-term pooling carries the full cloud (DPA / consent / minors) posture.
147     - **Compute** ? heavyweight VLM personalization (on-device Qwen3-VL fine-tune) is **out
of v1 scope**;
148     v1 is the cheap LogReg / threshold / cue-ON-OFF layer only.
149
150     ## 7. References
151     - `scratch/PERSONALIZATION_PIVOT_ANALYSIS.md` ? the decision-grade strategy analysis
(pros/cons, harness-fit, counter-case, recommendation).
152     - `scratch/PERSONALIZATION_FEATURE_DESIGN.md` ? the code-grounded design (schema,
serving bridge, APIs, critical files).
153     - Built: `backend/eval/per_user_gate.py` (#141), `backend/eval/per_user_eval.py` (#142).
154     - Harness: [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md), `backend/eval/experiment.py`,
`backend/eval/eval_stats.py`.
155

```

9. user (2026-06-15T14:58:06.064Z)

```

1     # Storage-priced sharing ? mental model (DRAFT v2 for co-design)
2
3     **Status:** DRAFT ? owner + Claude co-design session pending; nothing here is ratified.
4     **Date:** 2026-06-12 (v2 adds: model-tier axis, social layer, on-device end-state, abuse
economics ? per owner's second brief) . **Source:** owner briefs (voice note `Umiya
Velociti.m4a`
5
6     pending transcription) + UI_PROPOSAL . HOSTING_PLAN . PMF research . NEXT_STEPS quality

```

```

table.
7      ? If ratified, this **supersedes the monetization framing in COMMERCIALIZATION.md**
(away from
8      per-video/subscription, toward storage-anchored tiers that bundle model quality) ?
record as an ADR.
9
10     ## 1. The one-line model (v2)
11
12     Don't sell videos or subscriptions ? **sell shelf space, and let model quality ride the
shelf**:
13     free = **2 GB + the owned model** ; paid = **more shelf + the best model** ; the visible
meter is
14     always storage. The social layer makes the shelf worth filling; the owned model makes
the free
15     tier cost ~nothing; the end-state moves compute onto the customer's own hardware
entirely.
16
17     ## 1a. The three coupled axes (one user-visible meter)
18
19     | Axis | Free | Paid | End-state (serious users) |
20     |---|---|---|---|
21     | **Storage** (the meter) | 2 GB library | 10 GB+ / pooled | Mostly local disk; cloud =
shareable library only |
22     | **Model** (rides the tier) | Owned model only ? **never Gemini** | Best path (two-
stage refine / frontier) | Best owned model, on-device |
23     | **Compute** (the cost) | Our GPU, capped runs | Our GPU, priority | **Their GPU**
(USB-drive workflow) |
24
25     ## 2. The load-bearing decision: two tiers of object
26
27     | Tier | What | Size reality | Stored? | Quota? |
28     |---|---|---|---|---|
29     | **Source** (raw match upload) | Processing INPUT | 1?4 GB each | Temporarily: kept
through processing + correction window, then **auto-deleted** (e.g. 7 days or on "corrections
confirmed") | Not by default; optional "keep my raw match" toggle = counts |
30     | **Library** (compiled reels, kept rally clips, thumbnails) | The PRODUCT | reel ?
50?80 MB (3?5 min 720p) | Durable | **Yes ? this is what 2 GB means** |
31
32     Why this split: 2 GB cannot hold even one raw match, but holds **~25?40 reels** ? the
quota is
33     only meaningful against outputs. It also kills the "free CDN" abuse vector for a
storage-priced
34     product: **the library only ever contains pipeline outputs** ? there is no path to park
arbitrary
35     files in it. (Sources additionally pass the existing relevance validator.)
36
37     So: "**Do we store the uploaded video?" ? **Yes, transiently ? it's a processing input
with a
38     retention deadline, not a possession.** The reel is the possession.
39
40     ## 3. Eviction policies = a pluggable registry (the repo's own pattern)
41
42     `EvictionPolicy` ABC + registry, exactly like `segmenter_registry`:
43     `fifo` (oldest first) . `largest_first` . `smallest_first` . `least_visited` (view-
counter) .
44     `hard_stop` (no deletion ? block new saves until user frees space or upgrades).
45
46     - **Proposed default: `hard_stop`** ? never silently delete someone's match memories;
FIFO etc.
47     are opt-in conveniences. (Co-design question #2.)
48     - **Grace flow:** over quota ? banner + email with the exact list the chosen policy
would evict ?
49     N-day grace (propose 7) ? evict, logged in an `eviction_log`. Offer download-before-
evict.
50     - `least_visited` needs a decision: owner views, public views, or both?

```

```

51
52     ## 4. CUJ after upload (end-to-end)
53
54     ```
55     record ? upload source ? async job (PR-1) ? review/correct (corpus engine, PR-4)
56         ? compile reel(s) ? reel lands in LIBRARY (quota meter ticks)
57         ? share: profile page / unlisted link / private
58         ? source hits retention deadline ? auto-delete
59         ? library overflow ? eviction policy (with grace)
60     ```
61
62     A Gmail-style quota meter is permanent UI. Profile = `khelesutra.guru/@handle`: reel
63     gallery,
64     view counts (feeds `least_visited`), sport tags. Per-reel visibility: public / unlisted
65     / private.
66
67     ## 5. Upload path: "500 MB cap" vs "point at a bucket"
68
69     - Alpha (tunnel ? local box): keep the PR-2 chunked endpoint. A 500 MB cap ? 10?15
70     min of
71     1080p ? fine for pre-trimmed games, kills full matches (1?4 GB). Recommend keeping
72     the
73     4 GB chunked cap and rate-limiting per user instead; tomorrow's CUJ run will tell us
74     real file
75     sizes. (Co-design question #3.)
76
77     - Beta (hosted): the bucket IS the upload path ? browser ? object storage
78     directly via
79     signed resumable URLs (PLATFORM P2 `StorageBackend`); the server never proxies bytes;
80     "max
81     filesize" becomes policy enforced at URL-issue + post-upload validation, not a proxy
82     limit.
83
84     ## 6. The real bill is egress, not storage (sharing = bandwidth)
85
86     Storing a free user's 2 GB costs < ?5/month anywhere. Serving is the risk: on GCS,
87     egress is
88     ~$0.12/GB ? one mildly viral reel (80 MB × 1,000 views) ? $10 of bandwidth; on
89     Cloudflare R2,
90     egress is $0. ? Proposed hybrid: sources ? GCS (credits, lifecycle auto-delete
91     does the
92     retention for free) . library/serving ? R2 (zero egress; virality becomes free
93     marketing
94     instead of a bill). Storage-anchored pricing only has margins on zero-egress serving.
95
96     ## 7. Pricing sketch (INR, storage-anchored ? numbers are placeholders)
97
98     Free 2 GB ? paid tiers, e.g. +10 GB / +50 GB monthly or yearly. Anchors: Google One
99     is
100    ?130/mo for 100 GB (we are not selling bytes ? we sell organized, shareable match
101    memory* ?
102    priced above raw storage but inside the PMF envelope: B2C $60?180/yr). Academy (B2B
103    overlay):
104    pooled storage across students + coach-managed profiles. Processing cost note:
105    Gemini path is
106    ~$0.05/run ? decide whether the free tier includes unlimited processing or N
107    matches/month
108    (quota is storage-anchored, but compute isn't free ? soft cap recommended).
109
110    ## 7a. The model-tier axis ? why "free never touches Gemini" is load-bearing
111
112    1. Free-tier economics: Gemini ? $0.05/run forever; the owned model is ~$0 marginal
113    on owned
114    GPU. A free tier on the owned model is sustainable at scale; one on Gemini bleeds.
115
116    2. The corpus flywheel becomes the business model: free users correct rallies (PR-4)
117    ?

```

```

97     golden corpus grows ? owned model climbs the quality table (heuristic 0.48 ? Gen-0
0.74 ?
98     target ? wrapper's 0.66 on multi-court) ? free tier literally improves every quarter
?
99     more users ? more corrections. The owned-model roadmap
(OWNED_MODEL_IMPLEMENTATION_PLAN) IS
100     the free-tier product engine.
101     3. **Independence & privacy:** free tier immune to provider 503 waves, price hikes, and
terms
102     drift; paid tier uses Gemini only on its paid/private tier per existing guardrails.
103     4. **Honest quality ladder (from NEXT_STEPS):** free = owned (today Gen-0-grade,
improving);
104     mid = two-stage WASB+refine (0.83 clean); top = best available path (wrapper 0.93
clean).
105     Paid stays ahead until the owned model overtakes ? at which point margins improve
again.
106
107     ## 7b. Tier sketch (names placeholder ? co-design)
108
109     | | **Rally** (free) | **Smash** (paid) | **Court** (academy/pro) |
110     |---|---|---|---|
111     | Library | 2 GB | 10 GB+ | Pooled across students |
112     | Model | Owned only | Best path + priority queue | Best owned, **on-device** |
113     | Processing | N matches/mo cap | Higher cap | Unlimited (their hardware) |
114     | Social | Full | Full | + coach overlay, athlete profiles |
115
116     ## 7c. Social layer ? "Facebook of (match) videos," scoped honestly
117
118     The shelf is worth paying for only if filling it is social. **Interactions are free;
storage is
119     the meter** ? never paywall the social loop itself.
120     - **v1 social (already in the alpha/beta plan):** profile `@handle`, reel gallery, view
counts,
121     share links. **v2 social (post-PMF, big build):** follows, reactions (badminton-
native: ?
122     "clean smash"), comments, tag-a-player (consent-gated), academy groups, "rally of the
week."
123     - Network effect is local-first: the same court's regulars find each other ? academy =
built-in
124     graph seed.
125     - ? Scope discipline: feed/moderation/notifications is a product in itself; it lands
AFTER the
126     alpha proves retention, not alongside it.
127
128     ## 7d. On-device end-state ? the USB-drive vision
129
130     Serious users plug a drive of match videos into **their** Windows/GPU machine; the app
processes
131     locally (this repo already *is* that pipeline) and syncs only reels + metadata to the
cloud
132     library/profile. Cloud cost for the heaviest users ? storage + social only.
133     - **The licensing guardrail becomes the product moat:** Gemini can never ship on-device;
the
134     owned model (Qwen3-VL base, Apache-2.0 / MIT-only deps) CAN. Commercial-clean
licensing was
135     the prerequisite all along.
136     - Staging: academy pilot boxes first (serious, GPU-equipped, exactly the owner's own
setup) ?
137     consumer Windows app later (needs the WSL-free Linux-native/DirectML detector path ?
the
138     detector ABC seam exists for this).
139     - Maps to PLATFORM P3 `local`/`byo_worker` compute backends ? productized.
140
141     ## 7e. Abuse economics (why "serious pay, abusers can't extract value")
142

```

```

143     Each axis is self-defending: **storage** ? library accepts only pipeline outputs (no
arbitrary
144     files), sources auto-expire; **compute** ? free tier = owned model (cheap) + N runs/mo
cap +
145     relevance validator rejects non-sport uploads; **bandwidth** ? R2 zero-egress serving +
signed,
146     revocable links (and the egress allowance baked into paid storage pricing per $6:
price/GB =
147     storage + ?·expected-views·reel-size·egress-rate + margin; on R2 the ?-term ? 0 and
becomes
148     margin); **social** ? report/takedown + rate limits. There is no tier where an abuser
extracts
149     more value than they pay for, and no free capability that costs us marginal money at
scale.
150
151     ## 8. Guardrails (carry-overs, non-negotiable)
152
153     Consent-at-capture for other players (Phase-2 recording rules) · **minors never on
public
154     profiles** and never in the training pool · report/takedown path on public reels · share
links
155     signed + revocable · training-data opt-in stays separate from storage/sharing consent.
156
157     ## 9. Schema deltas (when we build ? not before ratification)
158
159     `users(quota_bytes, eviction_policy)` · `library_objects(id, owner, video_id,
kind){reel,clip,thumb},
160     bytes, created_at, last_visited_at, visit_count, visibility, storage_ref)` ·
`eviction_log` ·
161     source uploads gain `retention_deadline`. `storage_ref` follows P2's `StorageRef`
(backend id + key).
162
163     ## 10. Co-design agenda (the decisions, in order)
164
165     **Object model & policies**
166     1. Ratify the **two-tier model** (sources transient / library durable)?
167     2. **Default eviction:** `hard_stop` (proposed) vs `fifo`? Grace days = 7?
168     3. Quota counts reels only, or also opt-in kept sources? (proposed: both, sources opt-
in)
169     4. `least_visited`: whose visits count?
170
171     **Tiers & pricing**
172     5. Ratify **"free never touches Gemini"*** (owned-model-only free tier)?
173     6. Free tier processing cap: N matches/month = ?
174     7. Tier names/sizes (Rally 2 GB / Smash 10 GB / Court pooled ? placeholders) + INR price
points.
175     8. Egress-baked pricing accepted ($7e formula), with R2-for-library hybrid despite GCP
credits?
176
177     **Upload & serving**
178     9. **Alpha upload cap:** 500 MB vs keep 4 GB chunked? (use the first real CUJ file
sizes)
179
180     **Social & end-state**
181     10. v1 social scope = profiles + share links + view counts ONLY (feed/comments post-
PMF)?
182     11. Minors on public profiles: propose **hard exclusion** in v1 (stricter than the
training-pool
183         rule ? public sharing is a different risk class). Ratify?
184     12. Profile namespace: `@handles` on khelsutra.guru?
185     13. On-device milestone placement: academy pilot boxes first, consumer app after the
WSL-free
186         detector path?
187
188     **Paper trail**

```

189 14. ADR to supersede COMMERCIALIZATION.md's subscription framing once 1?13 settle.
190

10. assistant (2026-06-15T14:58:15.285Z)

These two docs are critical to the privacy lens. Key findings:

1. ****The product uses cloud Gemini for inference today**** (CLAUDE.md line 3: "with paid Gemini as the cloud AI"). This means raw video leaves the user's machine and is sent to Google's cloud for processing. The PERSONALIZATION_PLAN explicitly warns (line 145-146): *****Privacy deferral ? do not advertise 'data never leaves your machine' until on-device ships; near-term pooling carries the full cloud (DPA / consent / minors) posture.***** This is a direct internal instruction that the doc's "private and local-first" / "your video stays yours" framing risks violating.

2. ****Free-tier videos ARE pooled into the baseline automatically**** (PERSONALIZATION_PLAN line 57: "free-tier videos ARE pooled into the baseline (consent baked into the free terms)"). This is a major tension with the doc's flat "opt-in, nothing is used without explicit permission" claim. The consent is bundled into free-tier terms ? which is arguably not the same as the explicit, affirmative opt-in the doc describes. Pricing isn't set, so free tier may not exist yet, but this is a real future-state contradiction.

Let me verify the minors rule wording and whether the "local-first" claim is anywhere validated, plus check how the existing field-associate flyer handles these same claims for consistency.

11. user (2026-06-15T14:58:15.904Z)

```
1      # Field & Product Outreach Associate ? KhelSutra (Sports-Tech, Bengaluru)
2
3      **Help a Bengaluru sports-AI product meet its first real users.**
4
5      | | |
6      |---|---|
7      | **Duration** | ~3?4 weeks, part-time (flexible; mostly afternoons/evenings when
academies are active) |
8      | **Location** | Bengaluru ? in-person visits to badminton academies |
9      | **Pay** | ? ___ per completed academy visit + travel reimbursement; recording sessions
(stretch goal below) paid extra ? ? ___ per session |
10     | **Reports to** | Avi (phone/WhatsApp: ___ ) |
11
12     ---
13
14     ## The product, in plain words
15
16     **KhelSutra** turns ordinary practice/match video from a badminton academy into
17     ready-to-use pieces: it finds the rallies in the recording automatically, cuts
18     **highlight reels**, and produces a **simple summary** (how many rallies, how much
19     actual play time). It works on video from a phone on a tripod ? no special
20     cameras, no hardware to install. We're building a private web app that a few
21     invited academies will be able to try.
22
23     ## The job, in one paragraph
24
25     Visit good badminton academies across Bengaluru and have honest conversations with
26     **coaches AND students**: how do they work with video today, would something like
27     this actually help them, and **what would make it genuinely useful for them** ?
28     including what's missing or wrong with the idea. You work from a provided
29     conversation guide, write down what people **actually say** (the negative answers
30     matter most), and debrief with Avi after each visit. **You are gathering feedback,
31     not selling.** No tech background needed.
32
33     ## The stretch goal (extra pay)
34
35     For academies that are interested, you coordinate a simple recording session ?
36     a phone on a tripod, one-page checklist provided. We run that video through
37     KhelSutra, and you go back and show them **their own court, their own players, as
```



```

38 highlight reels and a report*: "this is what it looks like for you."* Then you
39 write down exactly what they say. There is no stronger feedback than that moment.
40
41 ## What you'll actually do
42
43 - Visit **10?15 academies** (intro + conversation, ~20?30 minutes each).
44 - Talk to the **owner/head coach** first, and ? with their permission ? a few
45 **adult (18+) students** about whether they'd watch/share their own rally
46 clips and what they'd want from them.
47 - Fill **one structured answer sheet per visit** (we provide it) ? their words,
48 not yours, including the most negative thing said.
49 - Show a short sample reel on your phone **only at the point the guide says to**
50 (after the open questions ? it changes people's answers if shown early).
51 - Debrief with Avi after each visit (WhatsApp/call). **Your first two visits are
52 done together with him**, so you're never improvising.
53 - *Stretch:* coordinate recording sessions and present the sample results back.
54
55 ## You're a good fit if
56
57 - You're comfortable walking into an academy and starting a conversation with a
58 stranger ? and just as comfortable hearing "no, this is useless to us."
59 - You can capture what people said **accurately** ? quotes, not summaries.
60 - Kannada / Hindi / English, as the conversation needs (you choose the mix).
61 - You play or follow a racket sport (helps a lot; not required).
62 - No technical skills needed. Reliability matters more than anything else.
63
64 ## What you get
65
66 - Paid per completed honest visit ? **not** per positive answer. A well-documented
67 "they hated it, here's why" is a fully paid, fully successful visit.
68 - Ground-floor view of a real sports-AI product finding its market; your field
69 notes directly shape what gets built next.
70 - Flexible hours, short commitment, clear scope.
71
72 ## Ground rules (the short version ? full version provided on joining)
73
74 1. Honesty over positivity. Never oversell, never promise features, prices, or dates.
75 2. **No recording of any person without written consent** ? and sessions involving
76 **minors are never recorded** for our use, period. In regular visits you record
77 nothing at all: notes only.
78 3. People's contact details stay between you and Avi.
79
80 ## How to apply
81
82 Message Avi at ____ with **"FIELD"**, one line about a sport you follow, and one
83 line about a time you had to get honest feedback out of someone. Next step: a
84 15-minute phone call with Avi.
85

```

12. user (2026-06-15T14:58:16.216Z)

```

1 # Hosting Plan ? khelsutra.guru
2
3 **Status:** PROPOSAL ? awaiting owner ratification.
4 **Date:** 2026-06-10 · **Constraints:** budget $10?25/mo · GCP credits available ·
DigitalOcean
5 acceptable · alpha backend = owner's GPU box behind a tunnel · 10?25 alpha testers ·
registrar = GoDaddy.
6 **Companion:** `docs/UI_PROPOSAL.md`.
7
8 ## 1. Recommendation in one line
9
10 **Alpha: free edge + your box** (Cloudflare DNS/Pages/Access/Tunnel ? FastAPI on the GPU
box) at
11 **$0/mo**, saving GCP credits; **Beta: GCP control plane + your box as pull-worker**

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(Cloud Run +
12     GCS + Firebase Auth, credits-covered), keeping the edge where it's free.
13
14     Rationale for the Cloudflare pieces despite the GCP preference: GCP has no free
equivalent of the
15     Tunnel + Access combo (IAP requires a load balancer, ~$18/mo standing cost), and the
alpha needs
16     exactly three things GCP charges for but Cloudflare gives free: a secure tunnel to a
home box with
17     a custom domain, email-allowlist auth (free ?50 users ? fits 10?25 testers), and static
hosting.
18     Credits are better spent at beta on compute/storage, which is where GCP is excellent.
19
20     ## 2. Alpha topology ($0/mo)
21
22     ```
23     tester browser
24         ? https
25         ?
26     Cloudflare edge (free): DNS + TLS + Access (email-OTP allowlist)
27         ??? app.khelsutra.guru ? Cloudflare Pages (React SPA; auto-deploy from GitHub)
28         ??? api.khelsutra.guru ? Cloudflare Tunnel (outbound-only `cloudflared` service on
the box)
29
30         ?
31         ?
32         FastAPI :8000 on the GPU box (Windows)
33         ? Gemini segmenter path (no GPU in request path)
34         ? async jobs (#16), SQLite WAL, output/ on disk
35
36     - **No inbound ports opened** on the home network; `cloudflared` dials out. Home IP
never exposed.
37     - **Auth:** Cloudflare Access in front of BOTH hostnames; backend trusts the
38     `Cf-Access-Authenticated-User-Email` header (verify the Access JWT for rigor) for per-
user scoping.
39     - **Upload limit:** free-tier proxy caps request bodies at **100 MB** ? SPA uploads in
?90 MB
40     chunks (UI_PROPOSAL B2). Match videos (1?4 GB) flow fine as chunks.
41     - **Box offline = service offline.** Acceptable for alpha; SPA shows a friendly
"processing engine
42     offline" banner (a `/api/health` ping). Beta removes this with the queue.
43     - **Costs:** Pages, Tunnel, Access, DNS = $0. Only the Gemini per-run cost
(~$0.05/match) and
44     domain renewal.
45
46     ## 3. Beta topology (GCP credits; ~$15?20/mo post-credits)
47
48     ```
49     browser ??? Cloudflare edge (free, unchanged) ??? app.khelsutra.guru (Pages)
50                                     ??? api.khelsutra.guru ??? Cloud Run
(FastAPI control plane, scale-to-zero)
51                                     ? jobs
table (Cloud SQL Postgres)
52                                     ? signed
resumable URLs
53                                     ?
54                                     GCS buckets
(uploads / reels, lifecycle ? cold)
55                                     ? pull
jobs, outbound-only
56                                     GPU box = first
`byo_worker` (PLATFORM P3)
57     ```
58
59     - Uploads go **browser ? GCS directly** (resumable signed URLs) ? no proxy body limit,

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no home
60     bandwidth for ingest; the box pulls only what it processes (WASB/owned-model jobs),
Cloud Run
61     handles Gemini-path jobs entirely in-cloud.
62     - Auth graduates to **Firebase Auth** (free; Google + email sign-in; India-friendly
phone OTP
63     later); Access allowlist retires.
64     - Post-credits estimate: Cloud Run ~$0?5 (free tier: 2 M req + 180 K vCPU-s/mo, scale-
to-zero) .
65     GCS 100 GB ? $2.60 + egress $0.12/GB (reel downloads ? watch this line) . Cloud SQL
smallest
66     ? $10 ? **~$15?20/mo**, inside budget. Egress lever if it grows: serve reels from
Cloudflare R2
67     (zero egress) while keeping everything else GCP.
68
69     ## 4. Alternatives considered
70
71     | Option | Monthly (alpha ? beta) | Pros | Cons |
72     |---|---|---|---|
73     | **Recommended hybrid** (CF edge + box ? +GCP) | **$0 ? ~$0 w/ credits** (~$15?20
after) | Cheapest at every phase; credits used where they matter; maps to PLATFORM P2/P3 | Two
vendors (CF + GCP) |
74     | **All-DigitalOcean** | ~$11 ? ~$17 ($6 droplet + Spaces $5: 250 GiB + **1 TiB egress
included**) | One simple vendor; Spaces' included egress is the best video-download deal; per-
second billing | No credits; no scale-to-zero; still needs a tunnel-equivalent to reach the box
(run `frp`/WireGuard on the droplet) |
75     | **All-GCP from day 1** | ~$18+ ? credits | Single vendor, credits immediately | LB+IAP
standing cost just to gate an alpha; no free tunnel to the home box; credits burn on plumbing |
76     | **All-Cloudflare** (Pages+Workers+R2) | ~$0 ? ~$5 | Zero egress on R2; cheapest
forever | FastAPI doesn't run on Workers (API stays on box/Cloud Run anyway); ignores GCP
credits |
77
78     ## 5. Domain setup ? GoDaddy ? Cloudflare DNS (one-time, ~15 min, you)
79
80     1. Create a free Cloudflare account ? **Add a domain** ? `khelsutra.guru` ? Free plan.
Cloudflare
81     imports existing records and shows **two nameservers** (e.g.
`xxx.ns.cloudflare.com`).
82     2. GoDaddy ? **My Products ? khelsutra.guru ? Manage DNS ? Nameservers ? Change ?
83     "I'll use my own nameservers"** ? paste the two Cloudflare nameservers ? Save.
Propagation:
84     minutes to 24 h. (Registration/renewal stays at GoDaddy; only DNS moves. Optional
later:
85     transfer the registration to Cloudflare Registrar for at-cost .guru renewals.)
86     3. Records (I manage after takeover): `app` ? Pages project; `api` ? `**Cloudflare API token** (you create: dash ? My Profile ? API Tokens) | Zone:DNS:Edit,
Account:Pages:Edit, Account:Access:Apps:Edit, Account:Tunnel:Edit ? scoped to this zone | DNS
records, Pages deploys (wrangler), Access policies, tunnel config |
94     | **GitHub repo** `khelsutra` (private) | Pages connects once (dashboard, you; build
root = `web/`) ? every push auto-deploys | I write the SPA in your Projects folder; commits
pushed = deployed |
95     | **On the box** (one-time, I hand you exact commands) | `cloudflared` installed as a
Windows service + `config.yml` | The tunnel itself |
96     | **GCP (beta only)** | A `khelsutra` project + service account (Cloud Run admin,
Storage admin, Cloud SQL admin) | Control-plane deploys |
97
98     Day-to-day flow after setup: I build ? commit ? push ? Pages deploys the SPA
automatically;

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99      API/tunnel/DNS changes via the scoped token. Secrets (Gemini key, Access policy) never
live in the
100      repo; `.env` on the box + Cloudflare/GCP dashboards.
101
102      ## 7. Go-live security checklist (alpha)
103
104      Access allowlist ON for both hostnames ? CORS narrowed to `https://app.khelsutra.guru` ?
rate
105      limit + per-user upload quota ? `security.allowed_video_root` set ? upload size cap +
extension
106      allowlist ? **rotate the Gemini API key** (NEXT_STEPS housekeeping ? it was shared in
chat) ?
107      confirm no router port-forwards ? weekly backup: SQLite DB + `output/` ? GCS bucket
(rclone
108      scheduled task; credits).
109
110      ## 8. Decisions needed from owner
111
112      1. Ratify the **hybrid recommendation** (CF free edge now, GCP credits at beta)? Or
force
113      single-vendor (DO / GCP) despite the cost notes above?
114      2. OK to move **DNS to Cloudflare** (registration stays at GoDaddy)?
115      3. Beta reel storage: **GCS (credits)** vs **R2 (zero egress)** vs **DO Spaces (1 TiB
included)** ?
116      can defer until beta Drop 2.
117      4. Approve the access grants in S6 so I can run the setup end-to-end.
118
119      *Ratified choices get recorded in `docs/archives/decisions/DECISIONS.md`.*
120
```

13. assistant (2026-06-15T14:58:25.594Z)

The hosting plan confirms the critical reality decisively. Both alpha and beta topologies route video through the **Gemini segmenter path** ? alpha sends it from the home box to Google's Gemini API; beta processes Gemini-path jobs **entirely in-cloud** on Cloud Run with uploads going **browser ? GCS directly**. So today, for invited alpha/beta users, the video is uploaded to a server (the owner's box via Cloudflare tunnel) and sent to Google's Gemini cloud API for processing. It is **not** running on the user's own machine, and data **does** leave to a third party (Google).

This is the central privacy-lens problem: the doc's repeated **"private and local-first," "your video stays yours," "built private and local-first"*** framing is not honest for the current alpha/beta product. The internal plan explicitly warns against exactly this (PERSONALIZATION_PLAN line 145: "do not advertise 'data never leaves your machine' until on-device ships"). On-device is a future end-state (STORAGE_SHARING_MODEL §7d), not today.

I have enough to assess every privacy/consent claim rigorously. Let me also note the field-associate flyer's minors wording for consistency ("sessions involving minors are never recorded for our use, period") versus the product doc's narrower "never used for model improvement."

Now I'll compile the structured review.

14. user (2026-06-15T14:59:10.656Z)

Structured output provided successfully

15. assistant (2026-06-15T14:59:22.750Z)

I've completed the consent/privacy/safety review and returned it via StructuredOutput.

The headline finding: the draft's repeated "private and local-first" / "your video stays yours" framing is **not honest for the current alpha/beta product**. Per the repo's own ground truth ? CLAUDE.md ("with paid Gemini as the cloud AI") and HOSTING_PLAN.md (alpha sends video from the box to the Gemini API; beta processes Gemini-path jobs entirely in-cloud with uploads going

browser?GCS) ? video is uploaded to a server and sent to Google's cloud for processing today. On-device/local is a future end-state. Critically, PERSONALIZATION_PLAN.md line 145 contains an explicit internal guardrail: "do not advertise 'data never leaves your machine' until on-device ships." The doc breaks that guardrail in front of real prospective customers, in two places (body + FAQ).

Two more must-fixes: the absolute opt-in promise ("nothing used without explicit permission") is overstated against the planned free-tier "consent baked into the free terms" pooling (PERSONALIZATION_PLAN decision 5), and a should-fix that the minors rule is stated more narrowly here ("never used for model improvement") than the company's own stricter posture elsewhere (never recorded for our use; never on public profiles).

Relevant files:

- C:/Users/avidu/Projects/badminton-highlight-indexer/docs/KHELSUTRA_PRODUCT_OVERVIEW.md (the draft)
- C:/Users/avidu/Projects/badminton-highlight-indexer/CLAUDE.md (Gemini-as-cloud-AI ground truth)
- C:/Users/avidu/Projects/badminton-highlight-indexer/docs/HOSTING_PLAN.md (confirms cloud processing in alpha + beta)
- C:/Users/avidu/Projects/badminton-highlight-indexer/docs/PERSONALIZATION_PLAN.md (the "do not advertise data-locality" guardrail; free-tier pooling)
- C:/Users/avidu/Projects/badminton-highlight-indexer/docs/STORAGE_SHARING_MODEL.md (minors/public-profile + retention guardrails)
- C:/Users/avidu/Projects/badminton-highlight-indexer/docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md (stricter minors wording, for consistency)